

Qatar-9b

R-2465

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_250214 · created 2026-08-23T01:34:52+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460720.75 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.124 +/- 0.06
Transit depth (Rp/Rs)^2	1.55 +/- 1.49 [%]
Orbital Inclination (inc)	89.73 +/- 1.69
Airmass coefficient 1 (a1)	0.82 +/- 0.46
Airmass coefficient 2 (a2)	0.16 +/- 0.89
Scatter in the residuals of the lightcurve fit is	55.67 %
Best Comparison Star	None
Optimal Aperture	3.23
Optimal Annulus	11.46
Transit Duration (day)	0.0748 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.124 ± 0.06 vs 0.1489 (deviation 0.42σ)
Duration fit vs published	0.0748 d vs 0.0779 d (deviation 0.69σ)
Mid-time O–C	27.1 min
Depth SNR	2.1
Residual scatter	55.67 %

Light curve

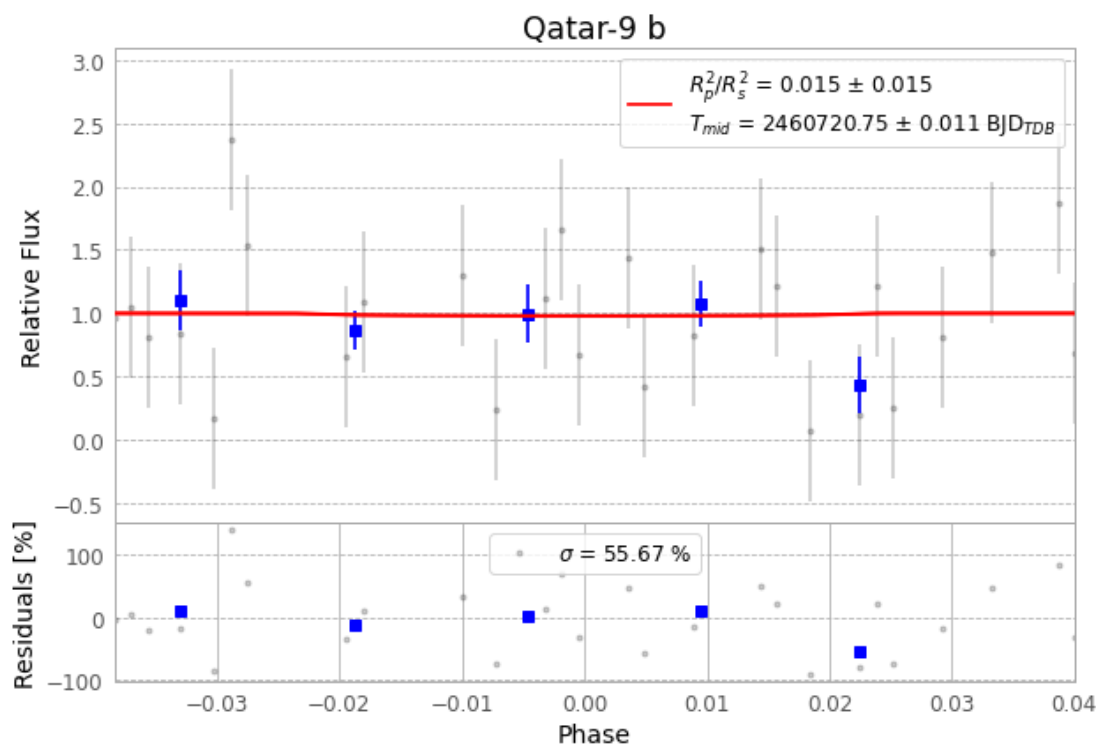


Figure 1 — FinalLightCurve_Qatar-9 b_2025-02-13.png (SHA-256 dbd447c2e4bd1293...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 92], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 eb20010a8016ebae...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

59 FITS frames (39 MB), Qatar-9250214042715.FITS → Qatar-9250214072116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-250214.log (6ba085ba5b53...), FinalLightCurve_Qatar-9 b_2025-02-13.png (dbd447c2e4bd...), FinalLightCurve_Qatar-9 b_2025-02-13.csv (c3513bf30713...)

Compute resources

Wall time 174.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 01:34Z.